



## UNIVERSITY OF DHAKA

Institute of Leather Engineering and Technology  
Department of Leather Engineering

# LABORATORY REPORT

Course Code: **LE-410**

Course Title: Wastewater and Solid Waste Management Lab

B.Sc. in Leather Engineering · 4th Year 2nd Semester · Examination 2025

### EXPERIMENT NO. 01

**Characterization of physiochemical parameters of tannery wastewater and comparison of identical water quality.**

|                         |                              |
|-------------------------|------------------------------|
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| Class roll number       | ZH-192-275                   |
| Examination roll number | 00                           |
| Registration number     | 2021115918                   |
| Session                 | 2021-22                      |
| Group number            | A(odd)ZH-192-275,269,039,005 |
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*Submitted to*  
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## DECLARATION OF ORIGINALITY

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**I declare that this laboratory report is my own work.**

I understand and accept the following:

- The experimental readings were obtained by my group, and the group's raw data and the pooled class data may legitimately be identical to those of my group members and classmates.
- Everything else in this report - the theory, the procedure as I have written it, the calculations, the discussion and the conclusion - must be written in my own words and represent my own understanding.
- Copying text from another student, from a previous year's report, or from any source without acknowledgement is plagiarism.
- All reports submitted for this course are compared with one another. Where the written sections of two or more reports are found to be substantially the same, all the students concerned will be penalised, whatever the source of the copying.

Scanned Signature of the student: MD. RIJU HOSSAIN Date: 2026-09-07

## 1. OBJECTIVE

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The objectives of this laboratory experiment are to:

1. Determine the pollution characteristics of the effluent by measuring parameters such as BOD, COD, TDS, TSS, chromium, sulfides, and chlorides.
2. Examine the possible environmental effects associated with the discharge of the effluent.
3. Develop a basic understanding of the design and functioning of an Effluent Treatment Plant (ETP).
4. Observe and assess the physical characteristics of the effluent, including its colour, odour, and turbidity.

## 2. THEORY AND PRINCIPLE

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Tannery wastewater refers to the liquid waste generated during the processing of animal hides and skins into leather. It may originate from operations such as soaking, liming, deliming, pickling, tanning, dyeing, and finishing. Among these stages, beamhouse operations generally contribute a major portion of the pollution load because of the extensive use of chemicals and the removal of organic and other unwanted materials from the hides.

The composition of tannery effluent depends on the processing methods and chemicals used. It may contain organic matter, salts, sulfides, chromium compounds, acids, and alkalis. These substances can affect the quality of receiving water and create environmental concerns if the wastewater is discharged without adequate treatment.

To assess the condition of tannery effluent, various physical and chemical parameters are determined. These measurements help indicate the degree of contamination, identify the nature of the pollutants, and assess whether the wastewater requires treatment before discharge.

The principal parameters considered in this experiment include:

- pH: Indicates the acidic or alkaline nature of the effluent.
- Temperature: Measures the thermal condition of the wastewater.
- Total Dissolved Solids (TDS): Represents the dissolved substances present in the sample.
- Total Suspended Solids (TSS): Measures suspended particles that can be separated from the wastewater.
- Total Solids (TS): Indicates the overall solids content of the effluent.
- Biochemical Oxygen Demand (BOD): Represents the oxygen required by microorganisms to decompose biodegradable organic matter.
- Chemical Oxygen Demand (COD): Measures the oxygen equivalent required to chemically oxidize oxidizable substances.
- Dissolved Oxygen (DO): Indicates the amount of oxygen available in the wastewater.
- Electrical Conductivity: Provides an indication of the concentration of dissolved ionic substances.

- Chloride: Measures the chloride content of the effluent.
- Sulfide: Determines the concentration of sulfide compounds present.
- Chromium: Measures the chromium content, which is particularly important in tannery wastewater because of its association with tanning operations.

The results obtained from these analyses can be used to evaluate the pollution characteristics of the effluent and understand the need for appropriate wastewater treatment. The laboratory determination of these parameters is generally carried out by following recognized procedures, such as those described in the APHA Standard Methods for the Examination of Water and Wastewater.

★ **pH** is a measure of the **acidity or alkalinity of an effluent**.

★ **TDS** represents the total amount of **dissolved inorganic and organic substances** present in water or effluent, such as salts, minerals, and ions.

★ **TSS** is the amount of **solid particles suspended in effluent** that can be separated by filtration.

★ **Electrical conductivity (EC)** is the ability of water or effluent to **conduct an electrical current**. It mainly depends on the concentration of dissolved ions.

★ **BOD** is the amount of **dissolved oxygen required by microorganisms to biologically decompose biodegradable organic matter** present in wastewater under specified conditions.

★ **COD** is the amount of **oxygen equivalent required to chemically oxidize organic and other oxidizable substances** present in wastewater.

### 3. APPARATUS AND REAGENTS

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#### 3.1 Apparatus and instruments

1. Multiparameter
2. Beaker of 500 mL
3. Glass rod
4. Hand gloove

### 3.2 Reagents and materials

1. Waste water for multiparameter test.

## 4. PROCEDURE

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1. The wastewater/effluent sample was collected in a clean container.
2. The sample was mixed gently to obtain a representative sample without producing excessive bubbles.
3. The **multiparameter meter** was calibrated according to the manufacturer's instructions using the appropriate standard calibration solutions.
4. A sufficient amount of wastewater sample was transferred into a clean beaker.
5. The **pH probe** of the multiparameter meter was immersed in the sample, ensuring that the sensing portion was completely covered.
6. The reading was allowed to stabilize, and all the **value** were recorded.

## 5. OBSERVATIONS AND DATA

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### 5.1 Sample and conditions

|                                 |                                         |
|---------------------------------|-----------------------------------------|
| Sample / stream tested          | Soaking ,liming and chrome liquor       |
| Variable assigned to your group | P <sup>H</sup> , BOD, COD, TSS, TDS, EC |
| Date and time of sampling       | 2026-06-12                              |

|                                    |                   |
|------------------------------------|-------------------|
| <b>Temperature</b>                 | 89.22 °F          |
| <b>pH</b>                          | 7.57              |
| <b>Any other initial condition</b> | Turbid appearance |

## 5.2 Your group's data

Multiparameter reading of wastewater :

| <b>Parameter / Trial</b> | <b>Value</b>          |
|--------------------------|-----------------------|
| Conductivity             | 7624 $\mu\text{S/cm}$ |
| TDS                      | 3811 ppm              |
| Resistivity              | 0.0001 MQ.cm          |
| Salinity                 | 4.54 PSU              |
| Temperature              | 89.22 °F              |
| Pressure                 | 755.8 mmHg            |
| <b>pH</b>                | 7.57                  |
| <b>mV(ORP)</b>           | -26.4 mV              |
| <b>mV/div</b>            | 59.6 mV/div           |
| <b>DO</b>                | 0.0% DO               |

## 5.3 Pooled class data

Multiparameter reading of lime liquor:

| <b>Parameter / Trial</b> | <b>Value</b>            |
|--------------------------|-------------------------|
| Conductivity             | 9559 $\mu\text{S/cm}$   |
| TDS                      | 2218 $\mu\text{S/cm}^2$ |
| Do                       | 0.005 ppm               |
| Temperature              | 86.36 °F                |
| <b>pH</b>                | 8.87                    |

Multiparameter reading of soaking liquor :

| Parameter / Trial | Value                         |
|-------------------|-------------------------------|
| Conductivity      | 908 $\mu\text{S}/\text{cm}$   |
| TDS               | 607 $\mu\text{S}/\text{cm}^2$ |
| Do                | 0.0 ppm                       |
| Temperature       | 86.73 $^{\circ}\text{F}$      |
| <b>pH</b>         | 8.20                          |

**Table 3: Waste water quality of World Health Organization (WHO)**

| Sample / condition assigned | Result obtained                                          | Unit                        | Remarks                                                 |
|-----------------------------|----------------------------------------------------------|-----------------------------|---------------------------------------------------------|
| <b>pH</b>                   | 6.5-8.5                                                  | -                           | Important for water acceptability                       |
| TDS                         | < 600                                                    | ppm                         | Mainly affects taste                                    |
| TSS                         | -                                                        | ppm                         | Reduce disinfection efficiency and water clarity        |
| Cr                          | 0.05                                                     | ppm                         | Health-based limit to protect against chromium toxicity |
| $\text{Cl}^-$               | 250                                                      | ppm                         | Mainly affects taste                                    |
| Turbidity                   | < 5 NTU(desirable)<br>$\leq 0.5$ NTU before disinfection | NTU                         | Affects water appearance                                |
| Sulphate                    | 250                                                      | ppm                         | Cause unpleasant taste                                  |
| $\text{NO}_3^-$             | 50                                                       | ppm                         | prevents methemoglobinemia                              |
| $\text{F}^-$                | 1.5                                                      | ppm                         | Prevents dental and skeletal fluorosis                  |
| Lead                        | 0.01                                                     | ppm                         | Affects nervous system                                  |
| Arsenic                     | 0.01                                                     | ppm                         | Reduces the risk of cancer                              |
| Fe                          | -                                                        | ppm                         | Affects taste, color, staining rather than health       |
| E. coli                     | 0 per 100 mL                                             | CFU;<br>Colony forming unit | Indicates microbiological safety                        |



| Sample / condition assigned | Result obtained | Unit                  | Remarks                                 |
|-----------------------------|-----------------|-----------------------|-----------------------------------------|
| Color                       | -               | TCU; true color units | Have no visible color in drinking water |

**Table 3: Waste water quality of United Nations Industrial Development Organization (UNIDO)**

| Sample / condition assigned | Result obtained            | Unit              | Remarks                                                        |
|-----------------------------|----------------------------|-------------------|----------------------------------------------------------------|
| <b>pH</b>                   | 6.5-8.5                    | -                 | Controls chemical reaction affects aquatic life                |
| TDS                         | <1000                      | ppm               | High levels may affect water quality                           |
| TSS                         | <1000                      | ppm               | High level reduce water clarity                                |
| Temperature                 | 20-35 °C                   | °C                | Affect aquatic organisms                                       |
| DO                          | ≥ 5                        | ppm               | Necessary for survival of aquatic life                         |
| BOD                         | ≤30                        | ppm               | Measure potential oxygen depletion                             |
| COD                         | ≤250                       | ppm               | Measures inorganic pollutants                                  |
| Turbidity                   | <5 NTU                     | NTU               | Measure water cloudiness                                       |
| PM <sub>2.5</sub>           | ≤30                        | mg/m <sup>3</sup> | Significant respiratory health impacts                         |
| Noise level                 | 55 dB(day)<br>45 dB(Night) | dB                | Indicates environmental noise pollution                        |
| PM <sub>10</sub>            | ≤45                        | mg/m <sup>3</sup> | Coarse inhalable particles that may cause respiratory problems |

**Table 4: Waste water discharge parameter provided by ISO.**

| Sample / condition assigned | Result obtained | Unit | Remarks                                         |
|-----------------------------|-----------------|------|-------------------------------------------------|
| <b>pH</b>                   | 6.5-8.5         | -    | Controls chemical reaction affects aquatic life |
| TSS                         | 30-100          | ppm  | High level reduce water clarity                 |
| Temperature                 | 35-40 °C        | °C   | Affect aquatic organisms                        |
| Oils and Grease             | 10-20           | ppm  | Can form a surface film,reduce oxygen transfer  |
| BOD                         | ≤50             | ppm  | Measure potential oxygen depletion              |
| COD                         | 125-250         | ppm  | Measures inorganic pollutants                   |

**Table 5: Waste water discharge parameter provided by EU.**

| Sample / condition assigned | Result obtained | Unit | Remarks                                                     |
|-----------------------------|-----------------|------|-------------------------------------------------------------|
| <b>pH</b>                   | 6.5-9.0         | -    | Controls chemical reaction affects aquatic life             |
| TSS                         | ≤35             | ppm  | High level reduce water clarity                             |
| Temperature                 | 30-35 °C        | °C   | Affect aquatic organisms                                    |
| DO                          | ≥ 5             | ppm  | Necessary for survival of aquatic life                      |
| BOD                         | ≤25             | ppm  | Measure potential oxygen depletion                          |
| COD                         | ≤125            | ppm  | Measures inorganic pollutants                               |
| Total phosphorus            | 1-2             | ppm  | Excess phosphorus can cause eutrophication and algal blooms |

| Sample / condition assigned | Result obtained | Unit | Remarks                                                                     |
|-----------------------------|-----------------|------|-----------------------------------------------------------------------------|
| Total nitrogen              | 10-15           | ppm  | Excess phosphorus can cause eutrophication and reduce aquatic water quality |

**Table 6: Waste water discharge parameter provided by ECR -2023. ( My Group-3)**

| Sample / condition assigned | Result obtained | Unit |
|-----------------------------|-----------------|------|
| pH                          | 6-9.0           | -    |
| SS                          | 150             | ppm  |
| Oils and Grease             | 10              | ppm  |
| BOD                         | 30              | ppm  |
| COD                         | 200             | ppm  |
| Cr                          | 2               | ppm  |
| Cl <sup>-</sup>             | 2000            | ppm  |
| Phenols                     | 1               | ppm  |

#### 5.4 Qualitative observations

The wastewater was dark/greyish, turbid, and had an unpleasant odour. Suspended solids were visible, and some particles settled upon standing. No significant floc formation was observed before treatment.

## 6. CALCULATION

Data observed by multiparameter.

### 6.1 Formula used

No formula used.

## 7. GRAPHS/ IMAGES:

## 8. RESULTS

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| Parameter determined | Result obtained              | Standard / limit for comparison |
|----------------------|------------------------------|---------------------------------|
| Conductivity         | 7624 $\mu\text{S}/\text{cm}$ | 1200 $\mu\text{S}/\text{cm}$    |
| pH                   | 7.57                         | 6-9                             |
| Temperature          | 89.22 $^{\circ}\text{F}$     | 104 $^{\circ}\text{F}$          |
| TDS                  | 381 ppm                      | <600 ppm                        |

## 9. DISCUSSION

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The measured results demonstrate that the three tannery liquors possess distinct pollution characteristics. The pH values were 8.20 for soak liquor, 8.87 for lime liquor, and 5.84 for Cr liquor. This variation can be attributed to the different chemicals and processing conditions involved in their preparation.

The relatively alkaline nature of the lime liquor is associated with the use of lime during the liming operation. In contrast, the lower pH of the Cr liquor is consistent with the acidic conditions generally maintained during the tanning stage.

The samples also exhibited TDS values considerably higher than the applicable standards, indicating a substantial concentration of dissolved substances. If such wastewater is released into water bodies without adequate treatment, it may pose serious risks to aquatic ecosystems and human health.

Therefore, the findings highlight the importance of treating tannery wastewater before discharge. Appropriate measures may include pH correction, reduction of suspended and dissolved pollutants, aeration where necessary, and specialized treatment for chromium-containing effluent.

## 10. CONCLUSION

The experiment provided an overall understanding of the physicochemical characteristics of tannery wastewater and its potential pollution impact. Different tannery operations produce effluents containing varying amounts of organic matter, dissolved solids, salts, sulfides, and chromium. The determination of parameters such as pH, TDS, TSS, BOD, COD, DO, conductivity, chloride, sulfide, and chromium helps assess the quality and pollution level of the effluent. Therefore, proper characterization is essential for selecting suitable treatment methods before tannery wastewater is discharged into the environment.

## 11. PRECAUTIONS AND SAFETY

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- Lab coat, gloves, and safety glasses were worn while handling tannery wastewater and chemicals.
- The wastewater samples were handled carefully to avoid skin contact, splashing, and inhalation of unpleasant odours.
- The multiparameter meter probe was rinsed with distilled water between measurements to prevent contamination.
- The meter was handled carefully and kept away from unnecessary contact with water.

- Wastewater and used chemicals were collected and disposed of according to the laboratory's waste-disposal procedure; they were not discharged directly into the sink

## 12. REFERENCES

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1. World Health Organization (2022) guidelines for Drinking water Quality (4th ed, incorporating the first and second agenda). Geneva :WHO
2. The Environment Conservation Rules, 1997 (ECR'97), Schedule-12, Section I – Bangladesh Department of Environment, published via FAOLEX